DRAFT SAMPLE EXAMINATION MARKING KEY

2008/18673[v2]

INTEGRATED SCIENCE STAGE 3—ANALYTIC MARKING KEY

SECTION ONE: MULTIPLE-CHOICE

Question Number	Answer
1.	c
2.	a
3.	c
4.	c
5.	b
6.	c
7.	d
8.	b
9.	d
10.	b
11.	c
12.	a
13.	c
14.	a
15.	d
16.	c
17.	d
18.	d
19.	b
20.	b

SECTION TWO: SHORT ANSWER AND PROBLEM SOLVING

Question 1

(a)

Description	Marks
• pit depth is 140 m below the surface whereas the water table is only 3m below the surface • water will seep into the pit from the surrounding groundwater	1 1
Total	2

(b)

Description	Marks
• head pressure increases with depth • greater seepage of water as mine deepens	1 1
Total	2

(c)

Description	Marks
• surface water - reduces the amount of surface water in the area • groundwater - water table in the vicinity of the mine will be lowered	1 1
Total	2

(d)

(i)

Description	Marks
• deep rooted trees would survive but trees and shrubs with root systems of 3-4 m would die within the draw down area • vegetation relying on surface water would die within the draw down area	1 1
Total	2

(ii)

Description	Marks
Any three points: • animals depending on surface water for drinking would move elsewhere if able • less food available for herbivores • less energy available for higher trophic levels • whole ecosystem impacted from the bottom up	max 3
Total	3

(iii)

Description	Marks
• subterranean fauna would die if aquifers dry out or water levels reduce	1
Total	1

(e)

(i)

Description	Marks
• the water level in the pit will slowly rise to form a lake	1
Total	1

(ii)

Description	Marks
• normally the inflow and outflow rate would keep a lake fresh • if the outflow rate is low and the evaporation rate is high, then the water in the pit will become more saline	1 1
Total	2

(f)

Description	Marks
waste products may seep into the ground water causing environmental problems if not accounted for	1
Total	1

Question 2
(a)

Description	Marks
• large area to survey • use of maps already existing to reduce exploration cost • minimal environmental impact	1 1 1
Total	3

(b)

Description	Marks
Gravity: • the gravity field of the earth is caused by the earth's mass. The strength of the field is a function of the composition of the mass (iron-rich centre with silicate minerals near the surface) and the distance away from it's centre. Where we stand on the surface, the gravity field has a field strength, which we can measure (as weight) • geophysicists have created a reliable model of the field • Using an instrument called a gravimeter, a geophysicist can measure the strength of the gravity field over a buried ore deposit to test for its presence. • If the gravity readings show a departure from the model's prediction of the normal field strength, it may indicate an abnormally dense rock mass at depth. • The gravity data can be examined in detail to highlight the anomalous readings **or** **Magnetic:** • The Earth has a magnetic field • Many minerals have a trace of magnetism which can be measured • Where magnetic minerals are present in abundance in a rock, the rock can be measured and traced with sophisticated magnetometers. • Geophysicists measure the field strength and orientation on the surface and compare it with predicted values using a model. • When the measured data on the surface shows anomalous values, there may be rocks below the surface causing the anomalies. **or**	1 1 1 1 1 1 1 1 1 1

Electrical: • a single conductor cable is connected to a current source, • the cable has a number of earth probes coupled to it at spaced points • switching units are activated in sequence so that the probes inject current into the earth. • at locations spaced from the probes the resistivity is measured and recorded for each injection. • resistivity measurements are compared with anomaly models to find a match of patterns, indicating the presence, size and location of an anomaly.	 1 1 1 1 1
Total	5

(c)

(i)

Description	Marks
• systematic collection of small samples of sediment from the beds of streams or creeks • sediments may be sieved or panned to see if minerals are present	1 1
Total	2

(ii)

Description	Marks
small samples preferably taken from unweathered bedrock for analysis	1
Total	1

(iii)

Description	Marks
• soil core sample collected by hand or from a small drill hole • surveyed grid used to locate the samples for analysis	1 1
Total	2

(d)

Description	Marks
• back filling • surface rehabilitation	1 1
Total	2

(e)

Description	Marks
• survey of area to identify unique or threatened species, or natural communities • monitoring of abiotic factors eg water table • plans for revegetation	1 1 1
Total	3

Question 3

(a)

Description	Marks
• removal of saleable trees for lumber	1
• about half a metre of topsoil and overburden is removed and conserved for later rehabilitation	1
Total	2

(b)

Description	Marks
Any three of the following in the answer • forest is a complex ecosystem • forests have trees of varying age with holes and rotting timber whereas new trees do not • not only the trees are removed, but other vegetation and the organisms it supports • replanting trees does not bring back the diversity • forests contain trees, other vegetation, microbes and animals	max 3
Total	3

(c)

Description	Marks
Any two of the following in the answer • deep ripping of the subsoil reduces compaction before the topsoil is replaced • allows increased water infiltration and storage • mixing of soil layers • removal of soil layers, therefore not original soil	max 2
Total	2

(d)

Description	Marks
• improved drainage	1
• growing conditions less favourable	1
Total	2

(e)

Description	Marks
• groundwater contamination	1
Total	1

(f)

Description	Marks
Any three of the following in the answer • neutralises corrosive residue • less damaging to environment if leakage occurs • useful carbon dioxide sequestration (carbon credits) • simple process not using chemicals	max 3
Total	3

(g)

Description	Marks
• caterpillars	1
• beetles	1
Total	2

(h)

(i)

Description	Marks
jarrah tree → caterpillar → wasp → skink →chuditch → hawk	1
Total	1

(ii)

Description	Marks
• loss of energy at each level of the food web	1
• insufficient energy to support more levels	1
Total	2

Question 4

(a)

Description	Marks
• process by which plants and some other organisms take the energy from sunlight to store in chemicals	1
Total	1

(b)

Description	Marks
• chemical energy produced by photosynthesis is passed from producers to consumers when they are eaten.	1
Total	1

(c)

Description	Marks
• energy expended in growing a crop probably produces greenhouse gases	1
• greenhouse gases contribute to global warming and climate change	
• we must get more useful energy out of the system, than we used to create the	1
source of energy	1
Total	3

(d)

Description	Marks
• anaerobic digestion -absence of oxygen	1
• bacteria convert organic material into methane and carbon dioxide gas	1
• sources include sewage, animal manure, household waste (rubbish tip) -two examples	2
Total	4

(e)

Description	Marks
Advantages: • renewable • conserves fossil fuels • spillage less polluting than petroleum	max 1
Disadvantages: • energy required to grow crop (planting, fertilising, harvesting) produces greenhouse gases • low energy density means less energy per kilogram • crop land used to produce energy and not food • ethanol contributes to the breakdown of rubber and plastic parts in vehicles • more expensive to produce than petrol • still adds carbon dioxide	max 1
Total	2

(f)

Description	Marks
• heat produced in incineration of waste can be used to produce steam	1
• steam used to turn the blades of the turbine	1
• mechanical energy of the rapidly rotating turbine is transformed into electrical energy by the generator attached to the shaft of the turbine	1
Total	3

Question 5
(a)

Description	Marks
• an electric current is passed through water producing oxygen and hydrogen gas	1
• diagram showing energy source, cathode, anode,	3
Total	4

(b)

Description	Marks
Any two of the following in the answer • hydrogen is non polluting • portability • high energy source • can be converted back into electricity	max 2
Total	2

(c)

Description	Marks
• hydrogen fuel cell	1
• combines hydrogen with oxidising agent (oxygen) to produce energy and form water	1
Total	2

(d)

Description	Marks
• liquid versus gas - harder to transport	1
• availability	1
• requires reinforced steel containers for storage	1
• need knowledge to handle gas safely	1
Total	4

Question 6
(a)

Description	Marks
Power saving = 100 - 18 = 82 W	1
Total time = 4 x 60 x 60 x 365 = 5 256 000 seconds	2
Energy = P x t =82 x 5 256 000 = 4.31×10^8 J	1
Total	4

(b)

Description	Marks
Number of CFL's = 48×10^6	1
Total energy saving = $48 \times 10^6 \times 4.31 \times 10^8 = 2.07 \times 10^{16}$ J	1
Total	2

(c)

Description	Marks
Any four valid answers • turn off appliances at the switch • more energy efficient appliances when replacing appliances • restrict use of air conditioning (by dressing appropriately)/use moderate heating and cooling temperatures for air conditioning • take shorter showers and smaller baths • use cold water instead of hot water • insulation • other appropriate answer	max 4
Total	4

(d)
(i)

Description	Marks
• E-W axis to avoid low angle sun reducing radiant energy	1
• north facing windows with eaves to transmit radiant energy from the sun in winter and block sun in summer	1
Total	2

(ii)

Description	Marks
• insulation materials such as foam, padding, reflective foil to minimise heat transfer by conduction • internal solid materials , such as brick, concrete used to conduct heat away from air mass • double glazed windows to minimise heat transfer by conduction • reflective roof surfaces such as colorbond, zincalume to minimise heat transfer by radiation	1 1 1
Total	3

(iii)

Description	Marks
• verandahs/eaves reduces radiant energy on windows in summer • skylights minimise use of lighting • roof ventilation - heat transfer by convection	1 1 1
Total	3

(e)

Description	Marks
Any four valid answers • use insulation/reflective backed curtains over windows - heat transfer radiation • pelmets over windows- heat transfer by convection • foam seals around and under doors - convection • block off unused chimneys and ventilation ducts - convection • close windows and doors in extreme temperatures -convection • other appropriate answer	max 4
Total	4

SECTION THREE: EXTENDED ANSWER

Question 7.

Description	**Marks**
Metal reactivity: • extraction method depends on the position of the metal in the activity series • alumimium is more reactive and is harder to reduce than iron • alumimium cannot be extracted by carbon reduction so it is obtained by electrolysis • iron is less reactive and therefore can be obtained by carbon reduction • carbon reduction is a cheaper method • large quantities of iron are produced by carbon reduction	1 1 1 1 1 1
	6
Chemical processes involved: **iron:** • iron requires carbon reduction • occurs in a blast furnace at high temperatures • iron ore, coke and limestone added to the top of the furnace • hot air is blasted into the bottom of the furnace • the air burns the coke to form carbon monoxide • the carbon monoxide reduces the iron oxide to form the iron **alumimium:** • alumimium requires electrolysis • alumina is mixed with molten cryolite • this reduces the temperature needed to extract aluminium • electric current passed through the mixture • aluminium smelting requires large amounts of electrical energy • carbon anodes are oxidised forming carbon dioxide • metallic aluminium to be deposited at the carbon cathode base and walls • large electrical currents required to keep the electrolyte molten	max 4 max 4
	8
Worksite hazards: • high temperatures involved in both processes • physical worksite injuries e.g. falls, machinery, vehicles **iron:** • carbon monoxide poisoning a hazard in iron production **aluminium:** • huge currents used - danger of electrocution	1 1 1 1
	4
Energy requirements: • both have high energy requirements • aluminium requires more energy than iron **iron** • heat required to fuel blast furnace • coke is burnt **aluminium** • high temperatures required for processing requires energy input • very high electrical energy requirements for electrolysis to produce aluminium	1 1 1 1 1 1
	6
short term, local effects: **iron**	

• blast furnace produces slag • iron oxide dust escapes from blast furnace **aluminium** • fumes containing perfluorocarbons - toxic **global environmental impacts:** • both produce carbon dioxide which is a greenhouse gas • aluminium produces very small quantities of perfluorocarbons which are very powerful greenhouse gases • oxides of nitrogen and sulfur produced - acid rain	1 1 1 1 1 1
	6
Total	**30**

Question 8

Description	Marks
current energy sources (2005) compared with predicted energy sources in 2030 • energy consumption will increase • predictions depend on policies • 2030 (RS) increase by 5 647 Mtoe increase and this is 17 100 /11 453 x 100= 49.3% increase in terms of Mtoe • 2030 (AS) increase by 3 947 Mtoe increase and this is 15 400/11453 x100 = 34.5% increase in terms of Mtoe	 1 1 2 2
	6
fossil fuels compared with alternative energy sources • current energy sources 81% from fossil fuels • 2030 (RS) 82% from fossil fuels -1% increase in fossil fuels as a source • 2030 (AS) 76% from fossil fuels i.e. 5% decrease in fossil fuels as a source	 1 2 2
	5
carbon dioxide emissions and contribution to greenhouse effect • current sources (oil, coal, gas) 9 276 Mtoe • 2030 (RS) (oil, coal, gas) 14 000 Mtoe • 2030 (AS) (oil, coal, gas) 11 700 Mtoe • increase in fossil fuels • increases carbon dioxide emissions in both scenarios	 1 1 1 1 1
	5
advantages of using two (defined)of the other energy sources: **e.g. biomass/biofuels** • renewable/ can be grown • relatively carbon neutral - uses carbon dioxide in growth and releases carbon dioxide when burnt **e.g. geothermal, solar, tidal, wind** • does not produce greenhouse gases • non polluting **disadvantages of using other energy sources:** **e.g. biomass/biofuels** • allocation of agricultural land away from food • fuel input to plant, harvest and transport fuel **e.g. geothermal, solar, tidal, wind** • high infrastructure costs • not always located near population due to source • area required for infrastructure • noise pollution (wind) • environmental impacts (tidal, wind)	Max 4 (2 for each source) Max 4 (2 for each source)
	8
advantages of using nuclear power: • does not produce smoke or carbon dioxide, so it does not contribute to the greenhouse effect • reduces need to use fossil fuels for generation of electricity making them available for industrial purposes • cost is comparative to coal • produces huge amounts of energy from small amounts of fuel • produces small amounts of waste • reliable **disadvantages of using nuclear power:** • waste disposal • potential nuclear accident • cost of decommissioning power station	max 3 max 3
	6
Total	**30**

Integrated Science: Sample Examination Stage3
Mapping questions to content

Quest. No	The practice of science		Conceptual understandings							The impact of science	
	3A	**3B**	**3A Resource exploration and environmental implications**	**3A Mining methods and environmental impacts**	**3A Extraction processing and environmental impact**	**3B Energy production and use**	**3B Fossil fuels and climate change**	**3B Nuclear power**	**3B Sustainable energy use**	**3A**	**3B**
Section One—Multiple Choice											
1				✓							
2			✓								
3			✓								
4					✓						
5	✓										
6	✓										
7	✓										
8					✓						
9					✓						
10					✓						
11								✓			
12						✓					
13							✓				
14		✓									
15						✓					
16									✓		
17		✓									
18									✓		
19						✓					
20									✓		

Quest. No	The practice of science		Conceptual understandings							The impact of science	
	3A	**3B**	**3A Resource exploration and environmental implications**	**3A Mining methods and environmental impacts**	**3A Extraction processing and environmental impact**	**3B Energy production and use**	**3B Fossil fuels and climate change**	**3B Nuclear power**	**3B Sustainable energy use**	**3A**	**3B**
Section Two—Short Answer											
1	✓			✓						✓	
2			✓							✓	
3				✓	✓					✓	
4									✓		
5						✓			✓		
6						✓			✓		

Quest. No	The practice of science		Conceptual understandings							The impact of science	
	3A	**3B**	**3A Resource exploration and environmental implications**	**3A Mining methods and environmental impacts**	**3A Extraction processing and environmental impact**	**3B Energy production and use**	**3B Fossil fuels and climate change**	**3B Nuclear power**	**3B Sustainable energy use**	**3A**	**3B**
Section Three—Extended Answer											
1					✓					✓	
2						✓	✓	✓	✓		✓

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